

Public Ensemble Weights Outperform Six Retraining Attempts in Interactive Whole-Body Lesion Segmentation

A Negative-Results Report from a Single-GPU Participant in autoPET V

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Abstract. We report our participation in the autoPET V challenge on interactive lesion segmentation in whole-body FDG- and PSMA-PET/CT. Working on a single consumer GPU (RTX 3060, 12 GB), we retrained the challenge baseline six times under controlled changes: a repaired click-channel pipeline, an on-the-fly click trainer, a click-dropout trainer, a residual-encoder architecture, a two-model ensemble, and three inference-side variants. None of them reached the organizers' own baseline on the preliminary leaderboard. On the fold-0 validation split, evaluated with one evaluator on identical data, the official baseline scores 0.7556 against our best retrained model at 0.7733; once the number of cases entering each mean is equalized, the official baseline is +0.041 ahead, and our apparent advantage is an artifact of 16 lesion-free cases being excluded from our mean but not from theirs. We therefore submit a 10-fold ensemble of the publicly released LesionLocator weights (Apache-2.0), which the challenge rules explicitly permit. Beyond the submission, we document three measurement failures that cost us most of the challenge period: the preliminary test set contains five cases of which four are scored, so leaderboard differences below roughly 0.05 are noise; training pseudo-Dice, single-case Dice, and a local multi-case interaction simulator each failed to predict leaderboard order; and a genuine click-blindness bug turned out not to explain the gap it was blamed for.

Keywords: autoPET challenge · Interactive segmentation · PET/CT · Negative results · Reproducibility

1 Introduction

The autoPET V challenge (MICCAI 2026) targets interactive lesion segmentation in whole-body PET/CT. Unlike the fully automatic tasks of earlier autoPET editions, an algorithm is called repeatedly: at the first iteration it receives the CT and PET volumes with an empty scribble set, and at each subsequent iteration it receives one additional scribble derived from its own previous error, namely

a foreground scribble on an under-segmented region or a background scribble on an over-segmented one, whichever component is smaller. The evaluation loop runs five iterations per case and ranks submissions by the area under the Dice curve and the area under the detection-F1 (DMM) curve, weighted equally.

Two properties of this protocol drive everything that follows. First, the zero-click iteration is included in the area, so a model that is strong only when clicked is heavily penalized. Second, each iteration adds exactly one scribble, so by the last iteration the model has seen at most four. The regime is sparse interaction, not dense correction.

We participated with a single consumer GPU. This report is deliberately weighted toward what did not work: our six retrained models, the diagnostics that identified and repaired a real defect in our training data, and the measurement mistakes that let us pursue a wrong hypothesis for several weeks. Our final submission uses publicly released weights rather than our own training, and we state plainly why.

2 Methods

2.1 Data

Training data. We used only the data provided by the challenge: 1611 whole-body PET/CT studies (FDG and PSMA tracers) with lesion annotations and simulated scribbles. No external or private data were used for the models reported here. All training used nnU-Net fold 0 of the organizers’ `splits_final.json` (1288 training / 323 validation cases), which we verified reproduces the organizers’ own split: applying nnU-Net’s rule (sorting the case keys, then `KFold(5, shuffle=True, random_state=12345)`) to the 1611 case identifiers yields exactly the 1288/323 partition reported in the official training log.

Validation data. Fold 0 of the same split (323 cases). We did not perform a full 5-fold cross-validation: a single 1000-epoch run takes approximately three days on our hardware, and the challenge period did not allow five of them. All validation numbers below are therefore fold-0 numbers, and we report the number of cases entering every mean, for reasons that Section 3 makes clear.

2.2 Data pre-processing

The network input has four channels: CT, PET, a foreground-click map and a background-click map. Because our early models were effectively blind to the click channels, we reverse-engineered the organizers’ click encoding rather than assuming it. Regenerating the click maps from the released scribble JSON files and comparing them against the organizers’ pre-computed heatmaps gave a byte-exact match on all 1611 cases: 1038 cases with clicks (2076 files, zero mismatches, zero out-of-bounds coordinates, no values other than 1.0) and 573 cases whose scribble lists are empty, for which the official heatmaps are uniformly zero.

The encoding is therefore a single voxel set to 1.0 with no smoothing, with the JSON coordinate triple indexed directly into the NIfTI array in $[x, y, z]$ order. Normalization follows the official configuration: CT normalization for the CT channel and Z-score normalization for the remaining three, including the sparse binary click channels. We note that Z-score normalization of a channel whose standard deviation is of order 10^{-4} amplifies the click voxels by roughly three orders of magnitude relative to the image channels, whereas substituting no-normalization, as we did for our on-the-fly models, leaves the clicks at 1.0 where they are dominated by the image channels. This is a plausible contributor to the weak click response we measured in those models, although we did not isolate it experimentally.

2.3 Algorithm/model

Final submission. Our final submission is a 10-fold ensemble of the publicly released LesionLocator interactive weights [1], distributed under the Apache-2.0 license by the team that won the corresponding autoPET IV task. The architecture is a residual-encoder U-Net (ResEnc-L) with a 192^3 patch and Euclidean-distance-transform click channels. The challenge rules state that publicly available data and networks may all be used, and we make no claim of novelty for the network itself.

Two changes were required to run the released code inside the challenge container: extending the fold list to all ten folds, and disabling test-time mirroring. We additionally strip each checkpoint to the four keys the nnU-Net predictor actually reads (`network_weights`, `trainer_name`, `inference_allowed_mirroring_axes`, `init_args`), which reduces each fold from 820 MB to 391 MB and the ten-fold model directory from 8.2 GB to 3.9 GB. The stripped checkpoints are reloaded and verified before the image is built.

Models we trained ourselves. For the negative results we report six configurations, all derived from the challenge baseline (plain convolutional U-Net, six stages, patch $112 \times 160 \times 128$): (i) a retrained baseline after repairing the click channels; (ii) an on-the-fly click trainer that synthesizes EDT click channels from the ground truth at every iteration; (iii) a click-dropout trainer ($p_{\text{drop}} = 0.5$) intended to break the shortcut described below; (iv) the on-the-fly trainer on a residual-encoder (ResEnc-M) backbone; (v) a softmax-average ensemble of (ii) and (iv); and (vi) three inference-side variants of the official weights (`checkpoint_best`, halved sliding-window step, full test-time mirroring).

2.4 Data post-processing

None. The predicted label map is written at the original geometry with no connected-component filtering, no size threshold and no morphological operation.

Table 1. Fold-0 validation Dice, identical data and evaluator. Here n is the number of cases entering the mean. The reported means are not comparable; the right-hand column is.

Model	n	Reported mean	Lesion-bearing only ($n = 207$)
Official baseline	223	0.7556	0.8140
Ours (repaired clicks)	207	0.7733	0.7733

2.5 Training and test parameters

Our own training used the nnU-Net defaults that the organizers also used, which we read from the baseline’s `debug.json` rather than assuming: SGD with learning rate 0.01, Nesterov momentum 0.99, weight decay 3×10^{-5} , polynomial learning-rate decay, 1000 epochs of 250 iterations, batch size 2, patch $112 \times 160 \times 128$, foreground oversampling 0.33, and the standard nnU-Net spatial and intensity augmentations. A single run took approximately three days on an RTX 3060.

At test time the final submission uses a sliding-window step size of 0.5 with mirroring disabled. One iteration of the 10-fold ensemble takes 171.9s on our RTX 3060 (161.0s of prediction, about 12.9s per fold, plus roughly 11s of container startup), that is 859.5s for the five iterations of one case against the 15-minute budget. On the challenge hardware, which uses a faster GPU, we expect this to be shorter, but we did not measure it.

2.6 Github repository

Link to Github repository: <https://github.com/leaieaie/autoPETV>

3 Results

Fold-0 validation. Table 1 compares the official baseline weights with our best retrained model on the same 323-case validation split, using one evaluator, one nnU-Net version and one prediction path. The comparison is not the one we first believed. nnU-Net’s mean validation Dice excludes cases in which the union of prediction and ground truth is empty. Our click-dependent model predicts nothing on lesion-free cases and is excluded from all 116 of them; the official baseline produces a false positive on 16 of those cases and is therefore scored 0.0 on them. Equalizing the population reverses the ranking: restricted to the 207 cases with lesions, the official baseline is ahead by 0.041.

We also note that the mean validation Dice recorded in the organizers’ own training log (0.7225) differs from our re-measurement of the same weights (0.7556) by +0.033. Re-running our own model from raw data rather than from preprocessed data changed its score by 2.5×10^{-7} , so the discrepancy is not the prediction path; we attribute it to library or environment versions. Numbers taken from the official log should not be compared against numbers produced elsewhere.

Table 2. Preliminary test phase, AUC-Dice and AUC-F1. All values are means over four scored cases (see Section 4).

Submission	AUC-Dice	AUC-F1
Official baseline (reference)	0.7800	0.7554
On-the-fly click trainer	0.7525	0.6622
Click dropout ($p = 0.5$)	0.7161	0.6724
ResEnc-M on-the-fly	0.6805	0.7059
Two-model ensemble	0.7189	0.7263
Official weights, <code>checkpoint_best</code>	0.7709	0.7545
Official weights, step size 0.25	0.7781	0.7431
Official weights, full mirroring	0.7826	0.7461
LesionLocator, 6 folds	0.7334	0.6030

Preliminary leaderboard. Table 2 lists the submissions we made with models we trained ourselves and with the official weights. The official baseline, shown for reference, was not beaten by any of them. Our final submission, a ten-fold ensemble of the public LesionLocator weights, scored higher on the preliminary set than the six-fold variant listed here; we do not read anything further into that difference, for the reason given in Section 4.

Interaction behaviour. A six-step diagnostic on individual cases separates two distinct failure modes. Our first repaired model responds strongly to clicks, with the predicted volume rising from 847 to 20431 voxels over the iterations and a final-iteration DMM of 0.842, but it collapses at the zero-click iteration (Dice 0.071). The cause is a shortcut that is genuinely present in the data: of the 1611 training cases, 573 have empty scribble lists, and in a 300-case audit the correspondence between “no tumor clicks” and “empty label” was exact, with 189 cases with tumor clicks all non-empty and 111 cases without all empty. A model can satisfy the training objective by predicting nothing when no click is given. Click dropout repairs the zero-click iteration (Dice 0.071 to 0.861) but weakens the click response, and scored lower on the leaderboard.

4 Discussion

The preliminary phase cannot rank anything. The preliminary test set contains five cases, one of which is lesion-free and scored as null, so every leaderboard figure in Table 2 is a mean over four cases. Per-case metrics are downloadable from the evaluation page, and we should have looked at them on our first submission rather than our fifteenth. Testing the per-case differences between our best public-weight run and the official baseline gives $t = -0.39$: the 0.047 gap that we spent days interpreting is not distinguishable from zero. The same holds for the ± 0.005 differences among our three inference-side variants, which consumed three submission slots and three days. We report this because the ranking

on a four-case leaderboard is visible to every participant and invites exactly the over-interpretation we fell into.

Three predictors that do not work. We tried, in order, to select models by training pseudo-Dice, by single-case Dice, and by a local multi-case interaction simulator. Each was contradicted by the leaderboard. Pseudo-Dice preferred an epoch-992 checkpoint that scored 0.009 lower than the epoch-1000 checkpoint. Single-case Dice predicted the wrong direction for both inference variants we checked. The local simulator ranked our click-dropout model above our on-the-fly model; the leaderboard ranked them the other way. We do not have a working local proxy for this challenge’s metric, and we think this is the main reason our search was unproductive: without one, every hypothesis costs a submission slot.

A real bug that was not the cause. Our preprocessed click channels were entirely zero for every training case, a defect introduced by a data-preparation script that reported no errors. This is a genuine bug, it is fully documented, and repairing it changed the mean validation Dice from 0.4371 to 0.7733. It is nevertheless not the explanation for our leaderboard deficit, for two reasons: our on-the-fly trainer synthesizes its click channels during training and never reads the broken ones, yet still lost to the baseline; and every model trained after the repair also lost. Having found one sufficient-looking cause, we stopped looking and wrote it up as settled. The actual reason our retrained models trail the official baseline, trained on the same data, the same split, the same architecture and the same schedule, remains unidentified. Candidates we have not tested include library-version differences and the normalization of the click channels discussed above.

Why the area is dominated by the first iteration. Because the loop starts with no scribbles and adds one per iteration, AUC-Dice is largely a measure of fully automatic segmentation quality. A model tuned for responsiveness to interaction, as ours were, optimizes the part of the curve that carries the least weight. In hindsight the click-dropout direction, namely forcing the model to segment well with no clicks at all, was the right one, and $p_{\text{drop}} = 0.5$ was simply too aggressive.

On submitting public weights. Our final submission is not our own model. The challenge rules permit publicly available networks without qualification, and we prefer to state the situation accurately: on this task, with this data, a single consumer GPU and roughly three months, we did not produce a model that beats either the organizers’ baseline or the released weights of the previous edition’s winner. The contribution we can defend is the measurement record above, not the network.

5 Conclusion

We submit a 10-fold ensemble of publicly released Apache-2.0 weights for autoPET V, and report six retraining attempts that failed to reach the official

Table 3. Algorithm details.

Team name	Algorithm name (as submitted on grand-challenge)	Data pre-processing	Data post-processing
25plqki	nnUNet 3d fullres interactive + scribbles)	nnU-Net default re-None sampling; CT normal- ization for CT, Z-score for PET and both click channels; EDT click encoding	

baseline. The most transferable lessons are methodological: verify the encoding of auxiliary input channels against the organizers’ own artifacts before training on them; report the number of cases entering every mean, because nnU-Net’s exclusion of empty cases can reverse a comparison; and check how many cases a preliminary leaderboard actually scores before treating its ordering as information.

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